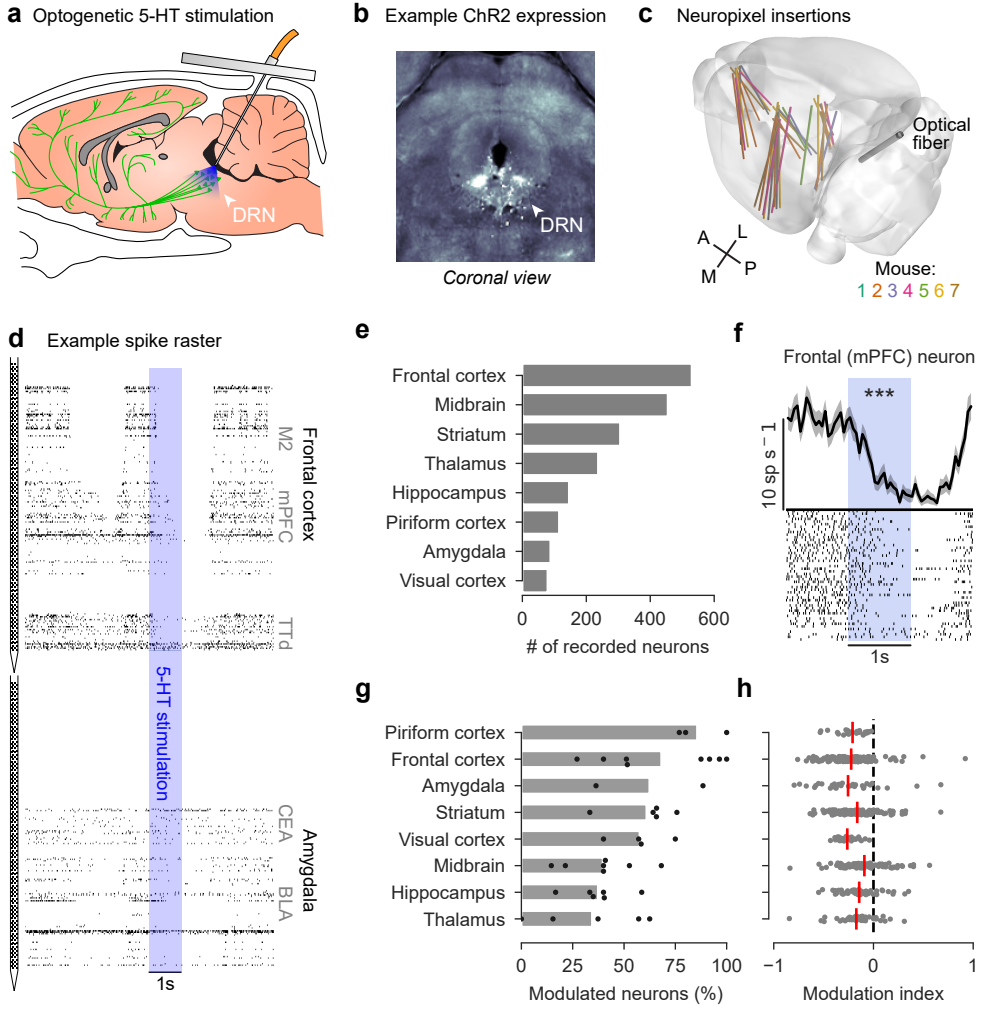

a
Optogenetic 5-HT stimulation
DRN
b
Example ChR2 expression
DRN
Coronal view
c
Neuropixel insertions
Optical fiber
A
L
M
P
Mouse:
1 2 3 4 5 6 7
d
Example spike raster
Frontal cortex
M2
mPFC
TTd
5-HT stimulation
Amygdala
CEA
BLA
1s
e
Frontal cortex
Midbrain
Striatum
Thalamus
Hippocampus
Piriform cortex
Amygdala
Visual cortex
0
200
400
600
of recorded neurons
f
Frontal (mPFC) neuron
***
10 sp s − 1
1s
g
Piriform cortex
Frontal cortex
Amygdala
Striatum
Visual cortex
Midbrain
Hippocampus
Thalamus
0
25
50
75
100
Modulated neurons (%)
h
−1
0
1
Modulation index